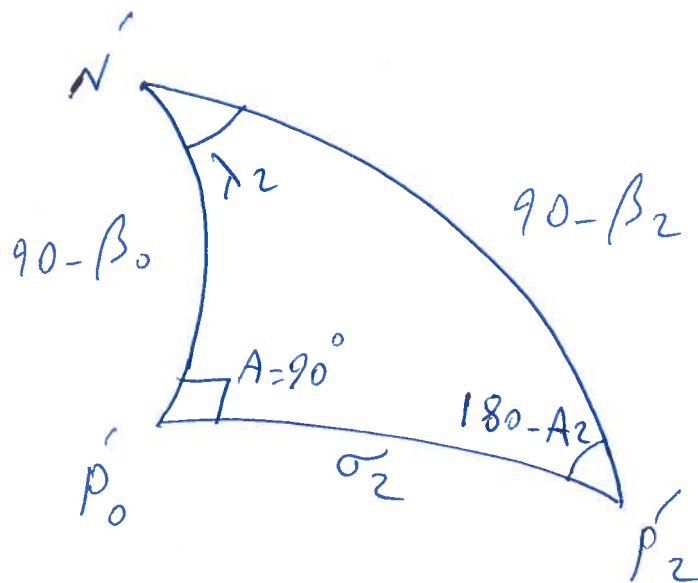




$$\gamma_{A_2} = \frac{-1}{\sin \sigma_2 \gamma \beta_0}$$

sin formula:

$$\frac{\sin 90^\circ}{\sin(90 - \beta_2)} = \frac{\sin(180 - A_2)}{\sin(90 - \beta_0)} \Rightarrow \frac{1}{\cos \beta_2} = \frac{\sin A_2}{\cos \beta_0} \quad (1)$$

five parts formula:

$$\cos(180 - A_2) \sin(90 - \beta_2) = \cos(90 - \beta_0) \sin \sigma_2 - \sin(90 - \beta_0) \cos \sigma_2 \underbrace{\cos 90^\circ}_{\text{zero}}$$

$$\Rightarrow \cos A_2 \cos \beta_2 = -\sin \beta_0 \sin \sigma_2 \quad (2)$$

$$(1) \Rightarrow \sin A_2 \cos \beta_2 = \cos \beta_0 \rightarrow \text{Divide by } \cos A_2 \Rightarrow$$

$$\Rightarrow \gamma_{A_2} = \frac{\cos \beta_0}{\cos A_2 \cos \beta_2} \rightarrow \text{From (2) we replace}$$

$$\Rightarrow \gamma_{A_2} = \frac{\cos \beta_0}{-\sin \beta_0 \sin \sigma_2}$$

$$\Rightarrow \gamma_{A_2} = \frac{-1}{\sin \sigma_2 \gamma \beta_0}$$